Report_DEB

1. Importation des données

```
file_path <- here::here("mod/DEB_SAFWI_dvw0")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```
NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 =
seeds <- c(C1 = 3333, C2 = 6666, C3 = 1212)

text_priors <- '
Distrib (pAm,          TruncNormal_cv,        100,      0.2,    40, 2000);
Distrib (Fm,           TruncNormal,            0.1,      0.2,    0.01, 0.8);
Distrib (r_pAm_pM,     TruncNormal_cv,         0.8,      0.1,    0.1, 2);
Distrib (kap,          Uniform,                0.1,     0.9);
Distrib (Ehp,          TruncNormal_cv,         100,      0.2,    40, 300);
Distrib (E_coc,        TruncNormal_cv,              30,      0.2, 1, 300);
Distrib (rOM_ClxHorse,  LogUniform,           0.00001,      0.3);
Distrib (f_Slope, TruncNormal_cv, 1, 1, 0, 10);
Distrib (dv, TruncNormal_cv, 2, 0.5, 0, 10);
Distrib (w0, LogUniform, 0.000000001, 0.1);
Distrib (f_EC, TruncNormal_cv, 0.5, 0.1, 0.1, 0.9);

# Calcul des vraissemblances
Distrib (Sigma_W, TruncNormal,  5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,       Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIte

# ./mcsim.DEBcaliSWI DEB_C1.in
```

`OM_horse` and `OM_soil` distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	pAm.1.	Fm.1.	r_pAm_pM.1.	kap.1.	Ehp.1.	E_coc.1.
Result_mode	205.907000	0.79082100	0.84282700	0.89982400	193.47600	9.7687100
2.5%	186.512000	0.71175265	0.69988300	0.89126100	162.38120	8.6723290
97.5%	221.879000	0.79919390	0.94288900	0.89992600	219.87300	11.3171000

```
Min          83.184900 0.04100250  0.63410200 0.27393000  63.11510  7.6467000
Max         238.066000 0.79999800  1.01875000 0.90000000 241.07800 42.4026000
Result_mean 204.361319 0.77411762  0.82440236 0.89756724 191.23574  9.8921445
Result_sd     9.060292 0.02435719  0.06272592 0.00558334  14.53425  0.7074828
            rOM_ClxHorse.1. f_Slope.1.     dv.1.        w0.1.    f_EC.1.
Result_mode    0.0089422300   1.580540 2.3427100 1.122280e-09 0.54790300
2.5%           0.0055940600   1.067560 1.9247200 5.409400e-09 0.47295050
97.5%          0.0116173350   2.652703 2.6591900 5.277570e-05 0.66116155
Min            0.0000340354   0.335638 1.7515500 1.122280e-09 0.38331600
Max            0.0671745000   4.529390 3.1106800 1.213750e-04 0.73783700
Result_mean    0.0086424054   1.691283 2.2682385 1.091966e-05 0.56646740
Result_sd      0.0015877159   0.405313 0.1893647 1.547688e-05 0.04804951
            Sigma_W.1.
Result_mode 0.12369800
2.5%        0.10926800
97.5%       0.14181835
Min         0.10080700
Max         5.19852000
Result_mean 0.12444372
Result_sd   0.04500194

Number of observations, n =  277
Number of parameters, k =  11
Log-likelihood, LL =  -10.46479
BIC =  82.79377


Potential scale reduction factors:

                Point est. Upper C.I.
dv.1.                 1.06       1.18
E_coc.1.              1.01       1.03
Ehp.1.                1.00       1.00
f_EC.1.               1.00       1.01
f_Slope.1.            1.02       1.06
Fm.1.                 1.00       1.00
kap.1.                1.01       1.01
pAm.1.                1.02       1.08
r_pAm_pM.1.           1.06       1.18
rOM_ClxHorse.1.       1.01       1.03
Sigma_W.1.            1.00       1.01
w0.1.                 1.03       1.09
```

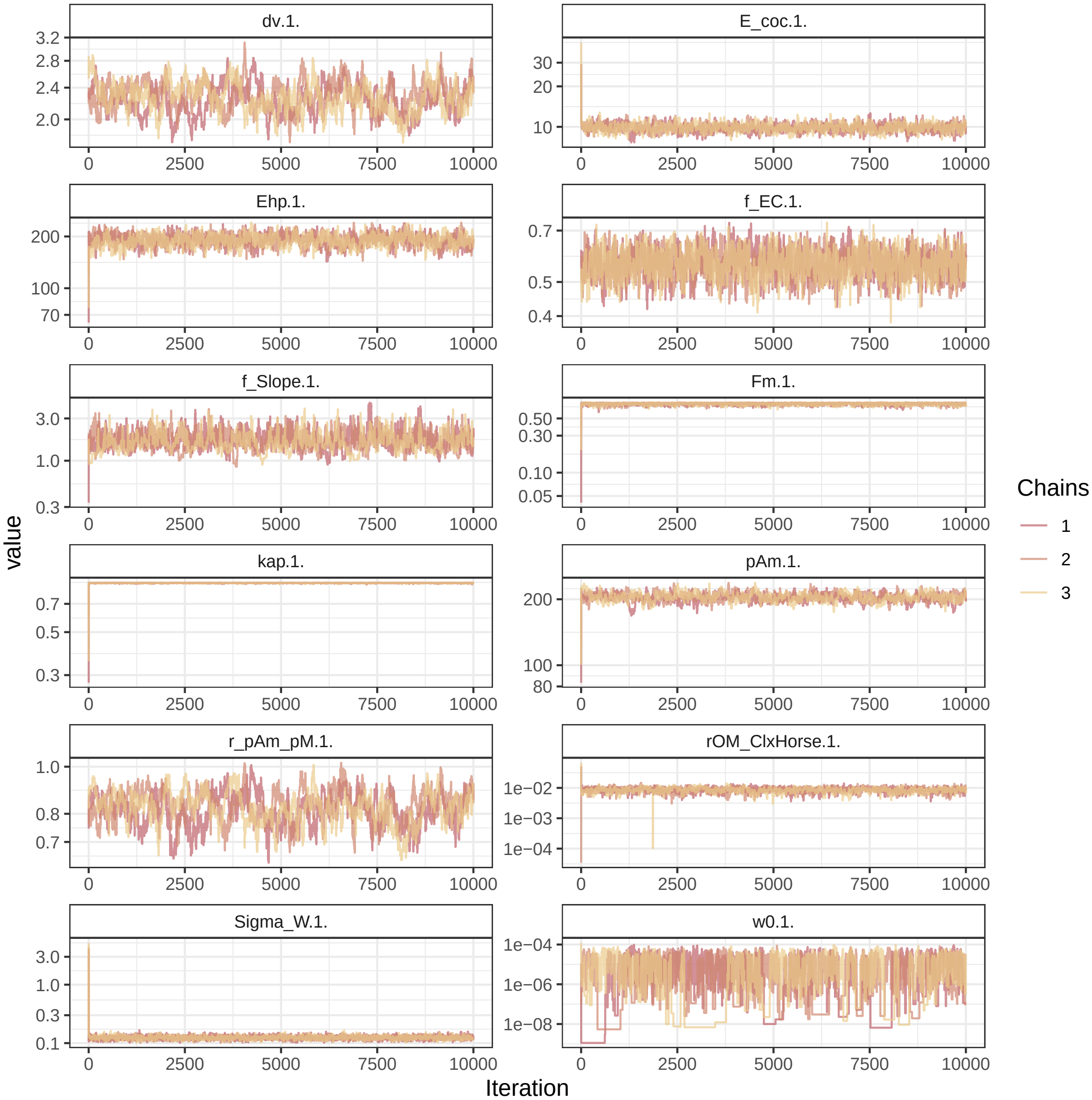


Figure 1: Parameters chains. Only the first iteration and the last `r Nb_iter_kept` are represented.

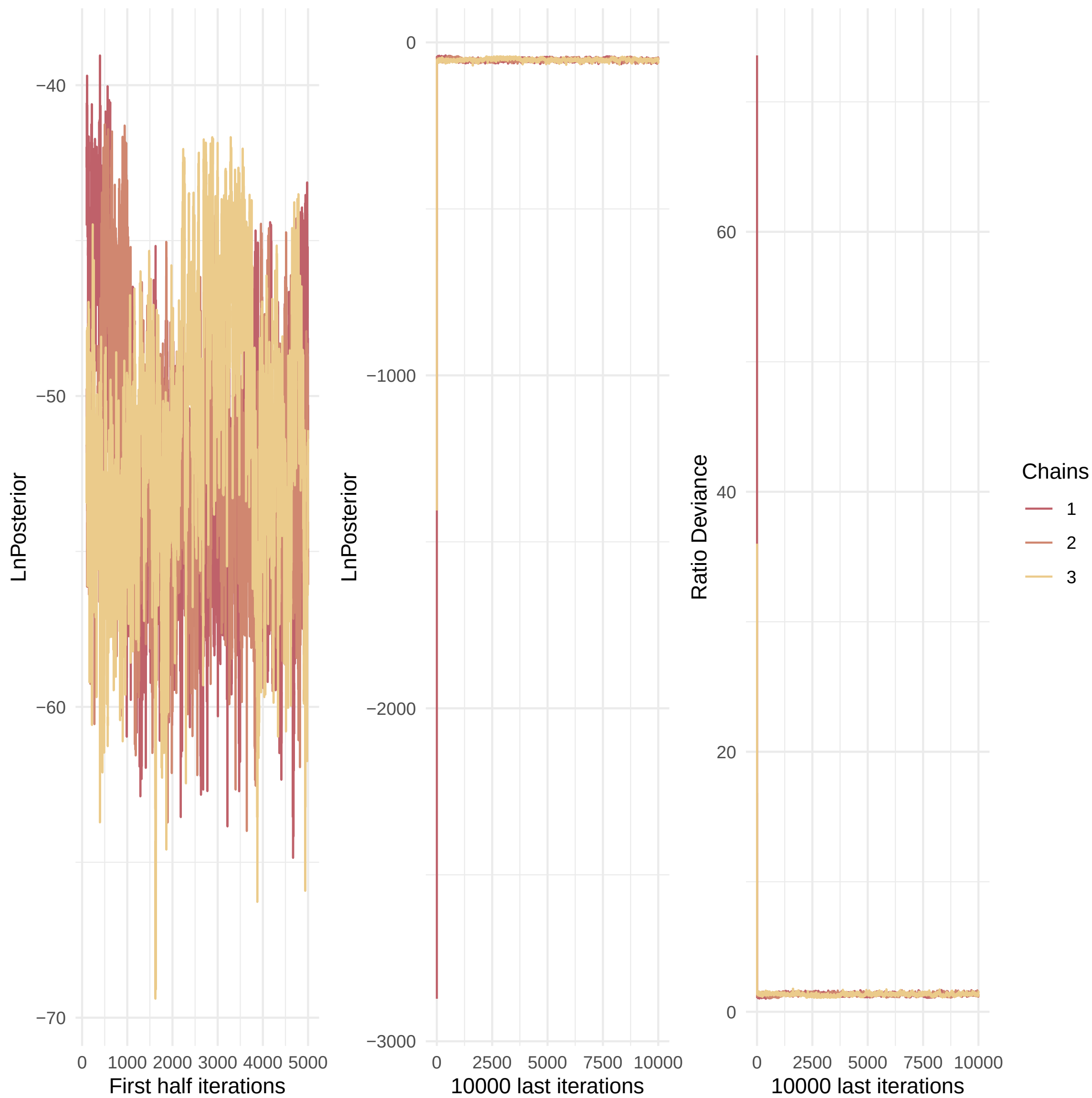


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

```
Multivariate psrf

1.06
```

```r
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE])  # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

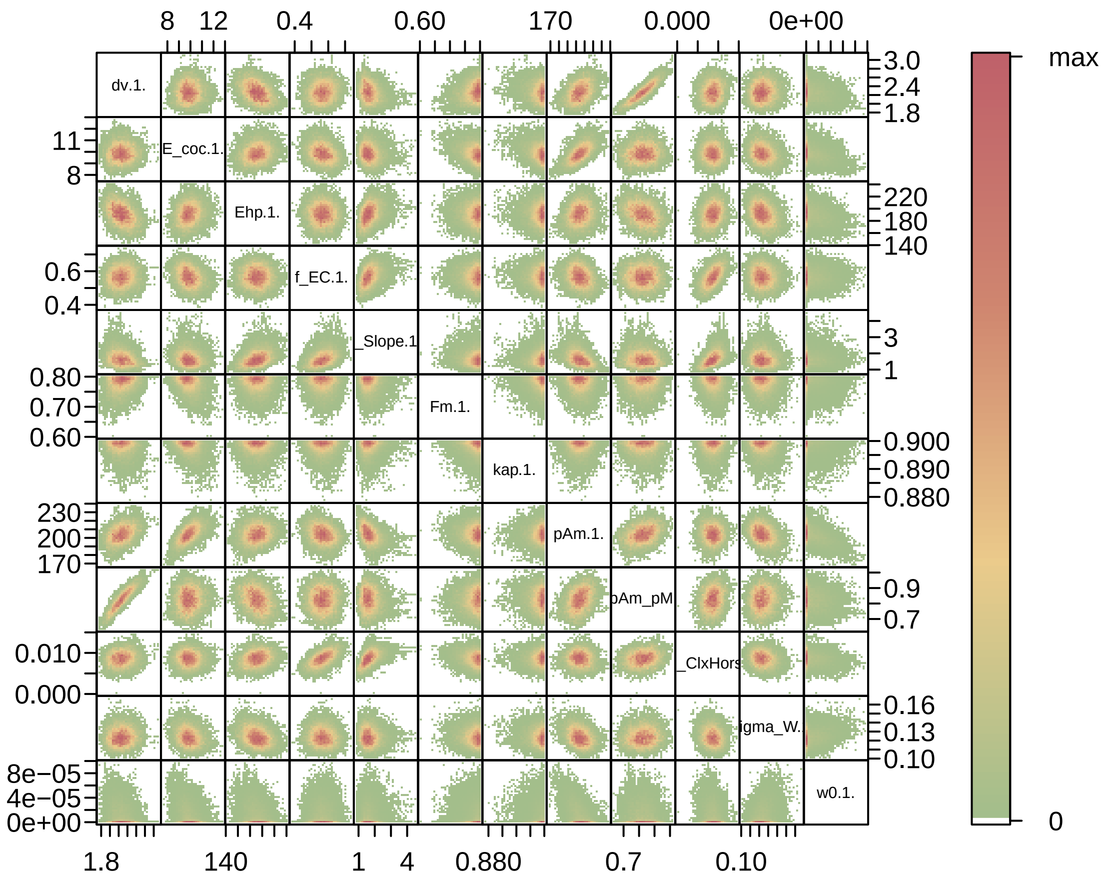


Figure 5: Parameter correlations.

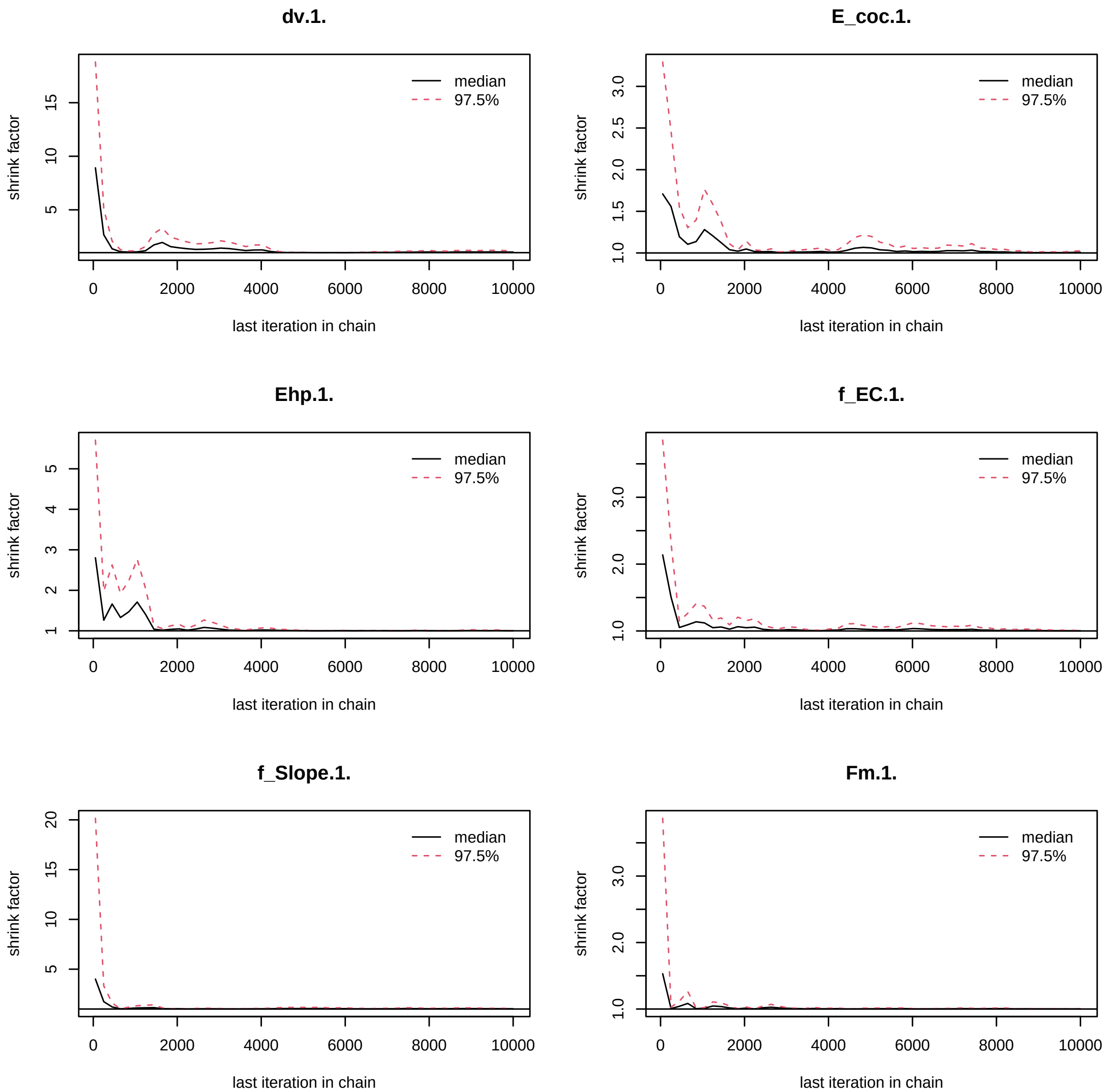


Figure 3: Chains convergence.

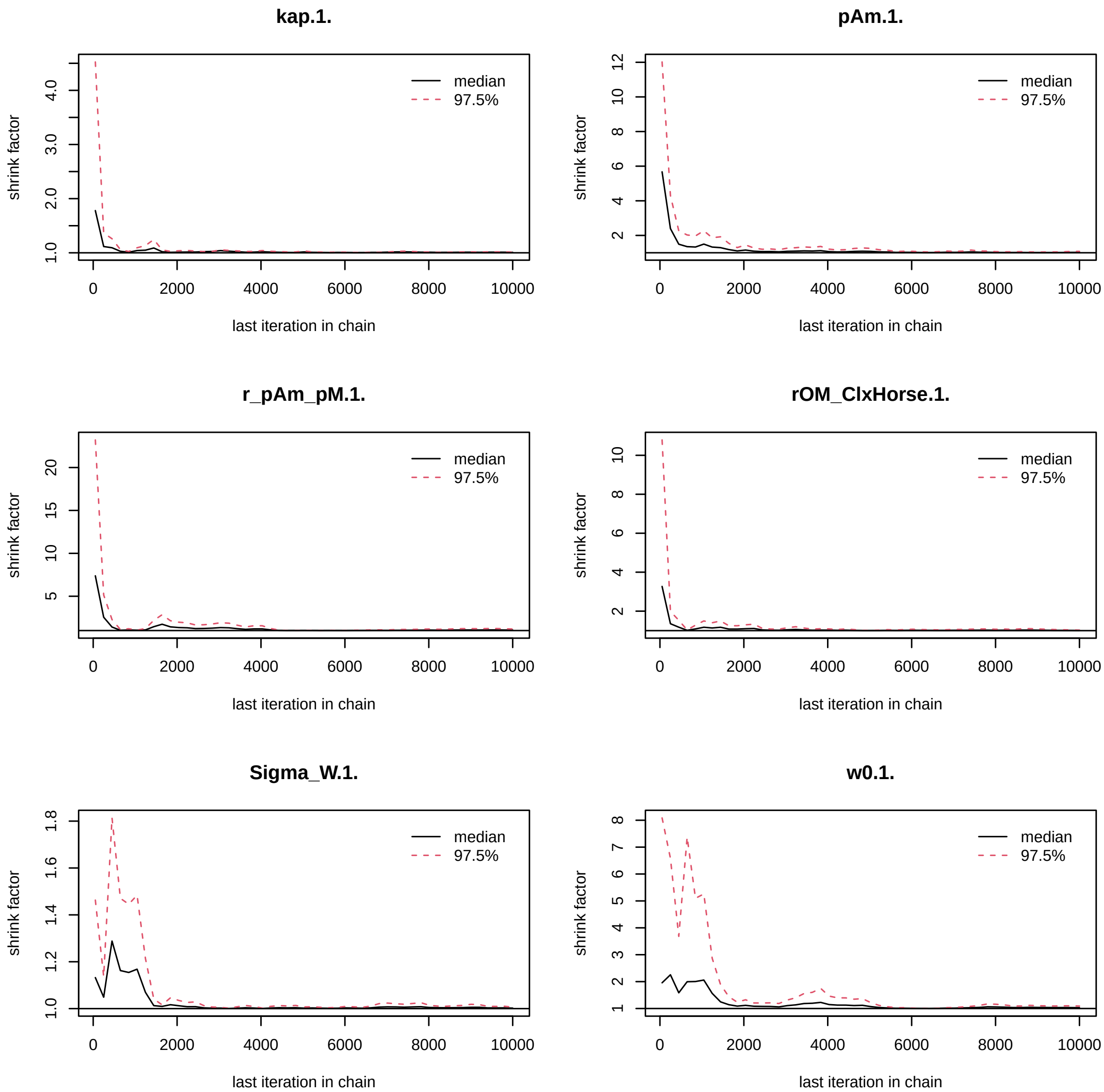


Figure 4: Chains convergence.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\.1\\.$|\\.1\\.", "", param),
    line  = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
    Length_struct,
    Reproduction,
    Energy,
    Organic_matter,"
```

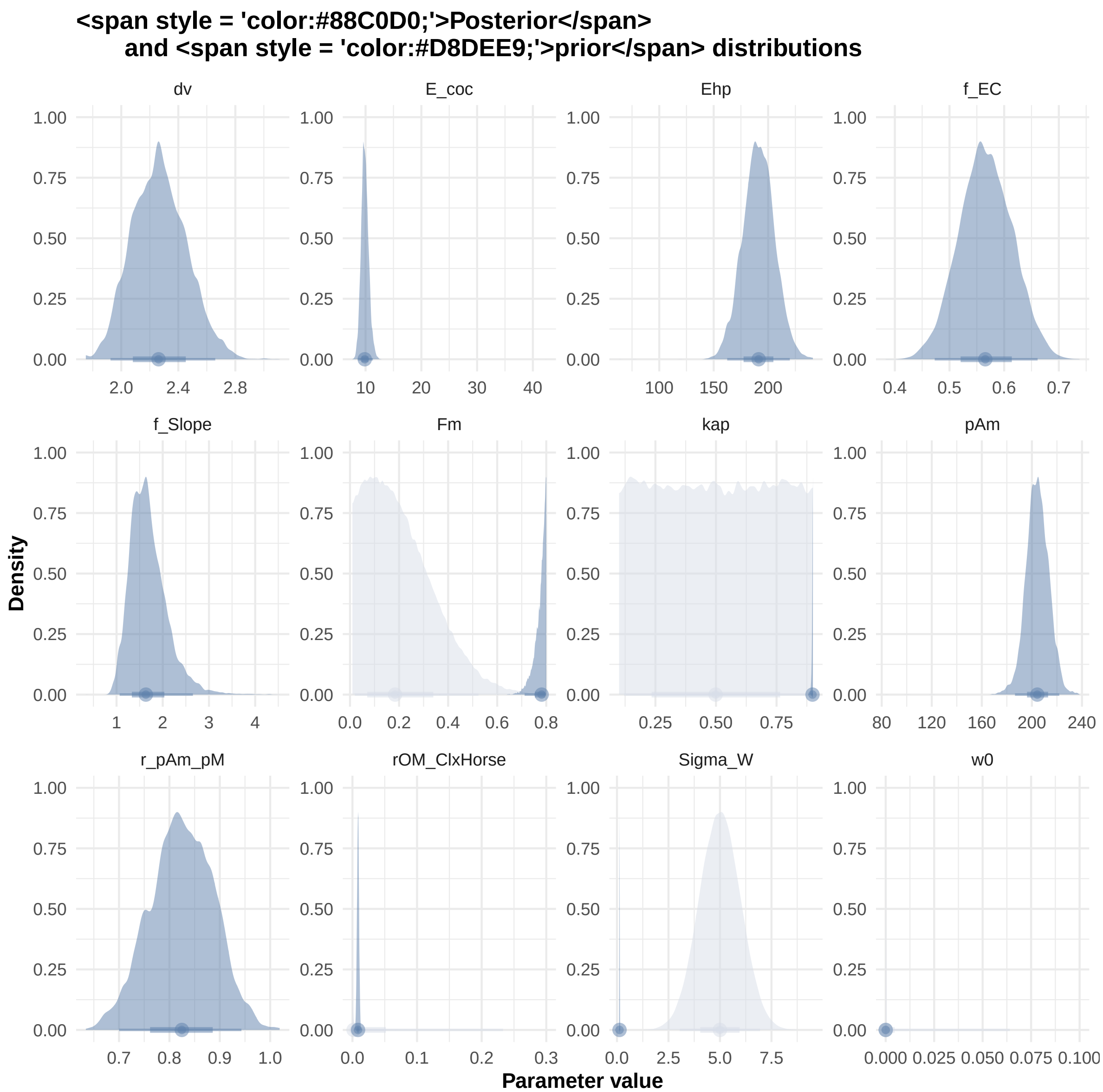
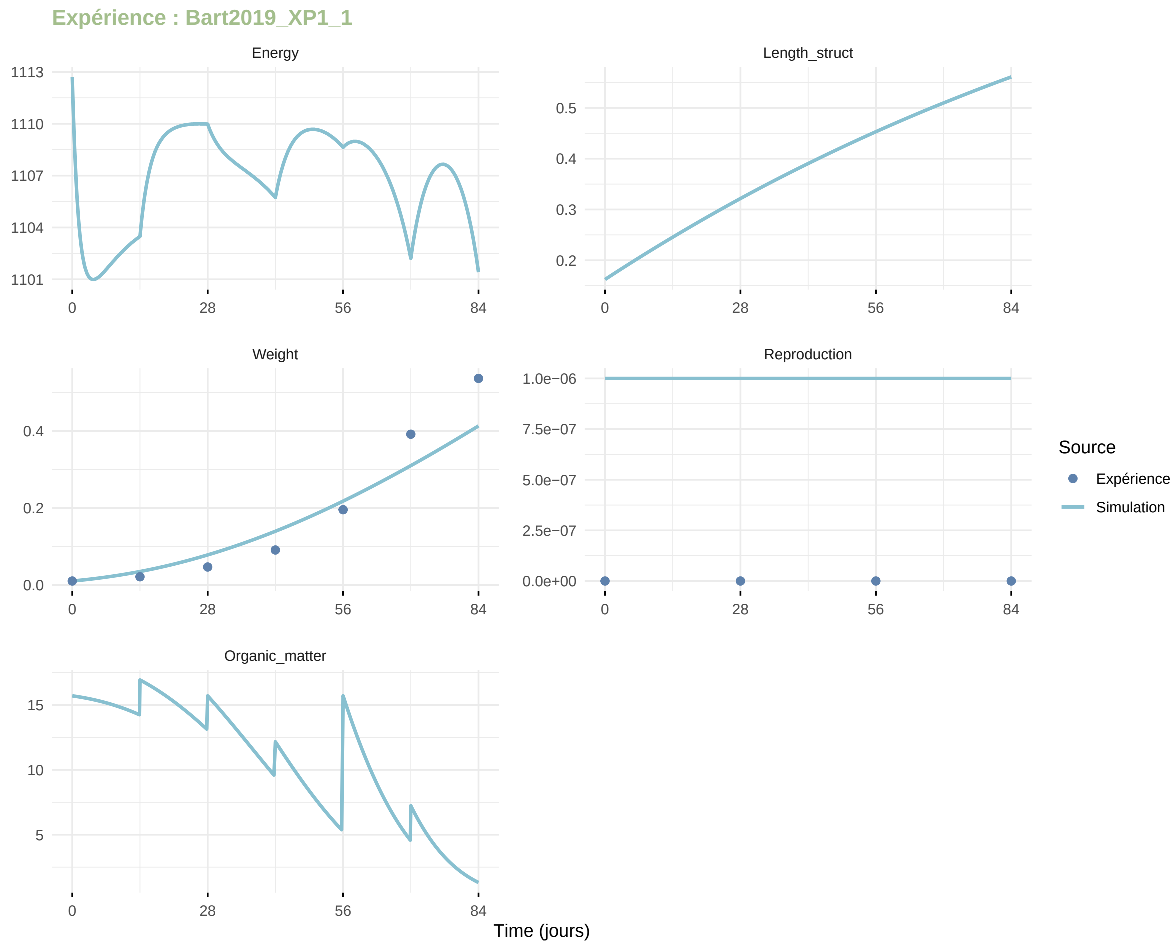


Figure 6: Distributions of the priors and the posteriors.

```
f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_prin
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pr

# ./mcsim.DEBcaliFWI DEB_sim_full.in
```

```
##############################
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#############################

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times)
```

```
Warning: `read_table2()` was deprecated in readr 2.0.0.
i Please use `read_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpe
i Row 1 of `x` matches multiple rows in `y`.
i Row 1 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
  "many-to-many"` to silence this warning.
```


Expérience : Bart2019_XP1_1
Energy
1113
1110
1107
1104
1101
0
28
56
84
Length_struct
0.5
0.4
0.3
0.2
Weight
0.4
0.2
0.0
Reproduction
1.0e−06
7.5e−07
5.0e−07
2.5e−07
0.0e+00
Source
Expérience
Simulation
Organic_matter
15
10
5
Time (jours)

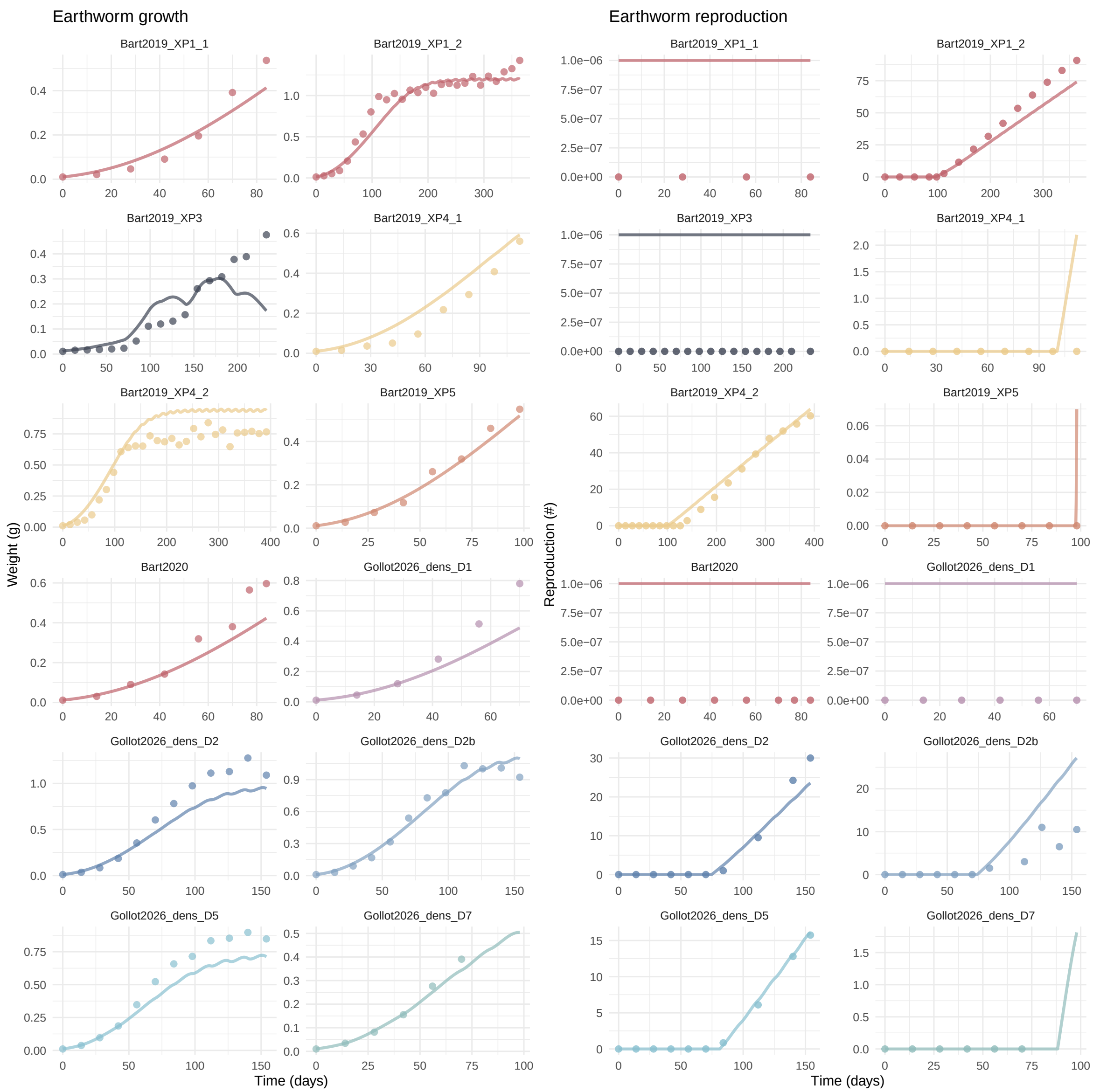
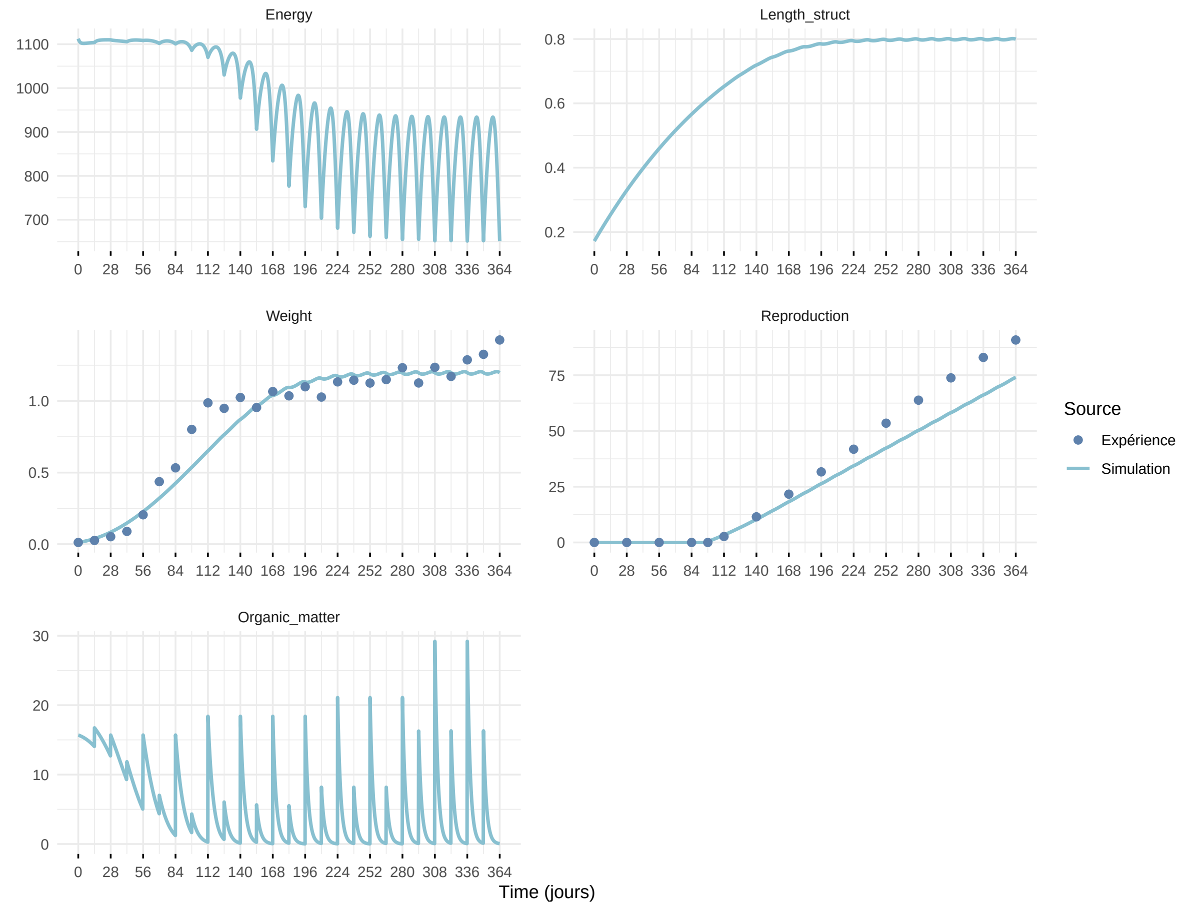
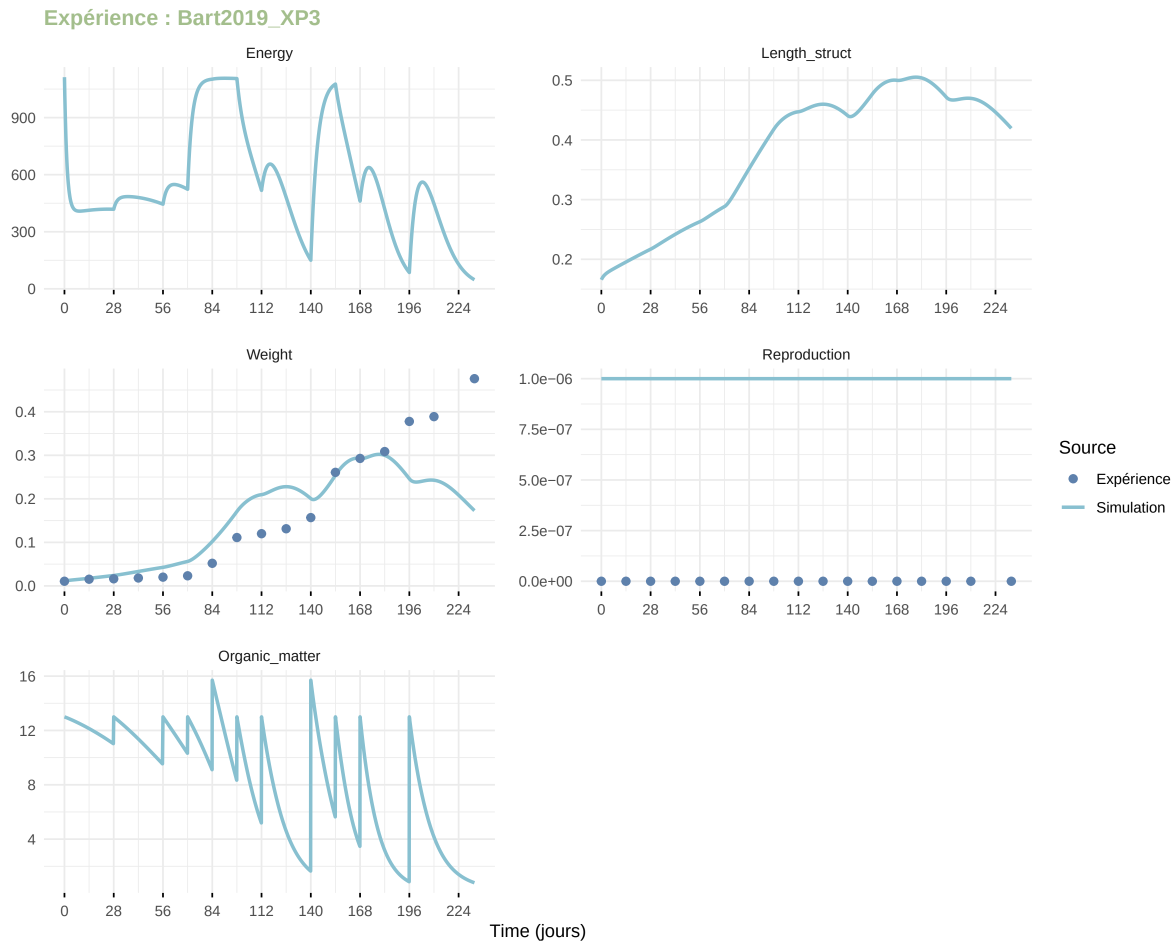


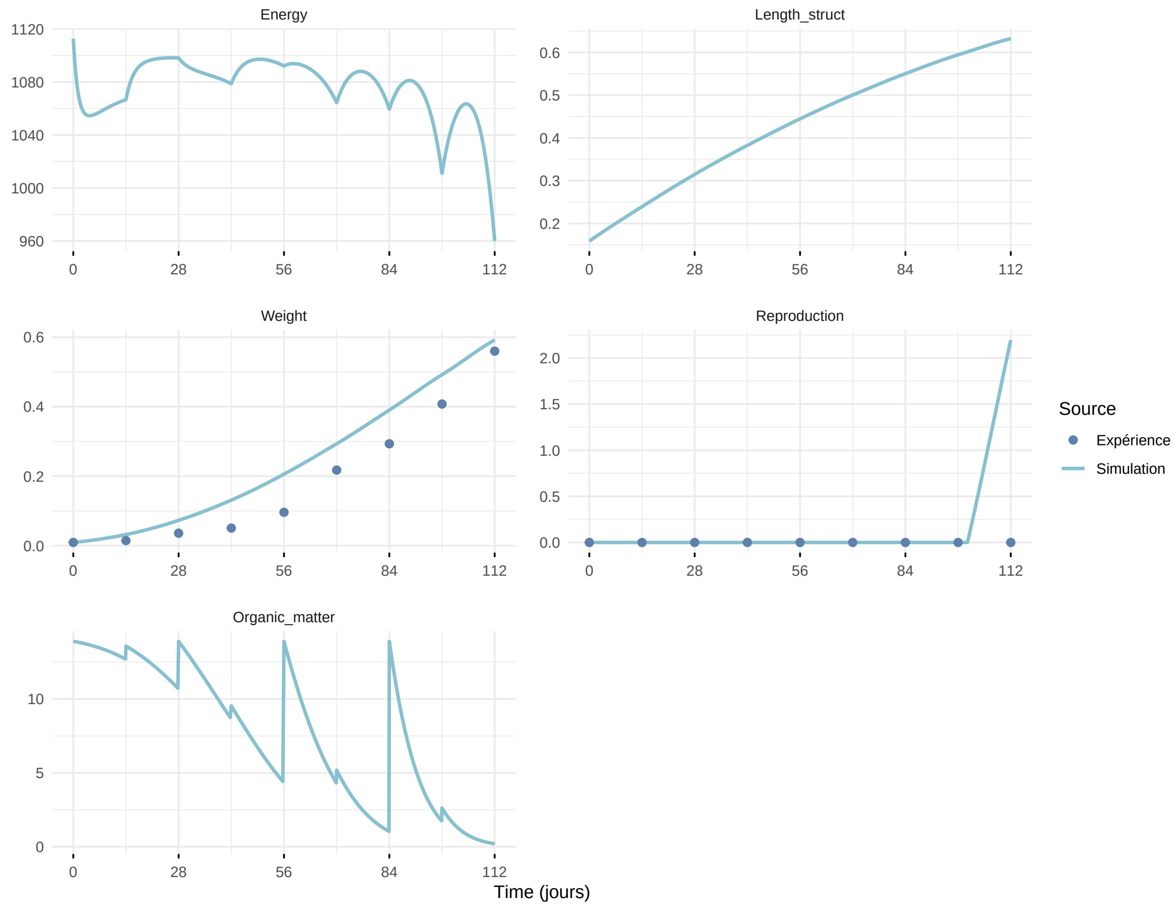
Figure 7: Simulations Weight and Reproduction.

Expérience : Bart2019_XP1_2

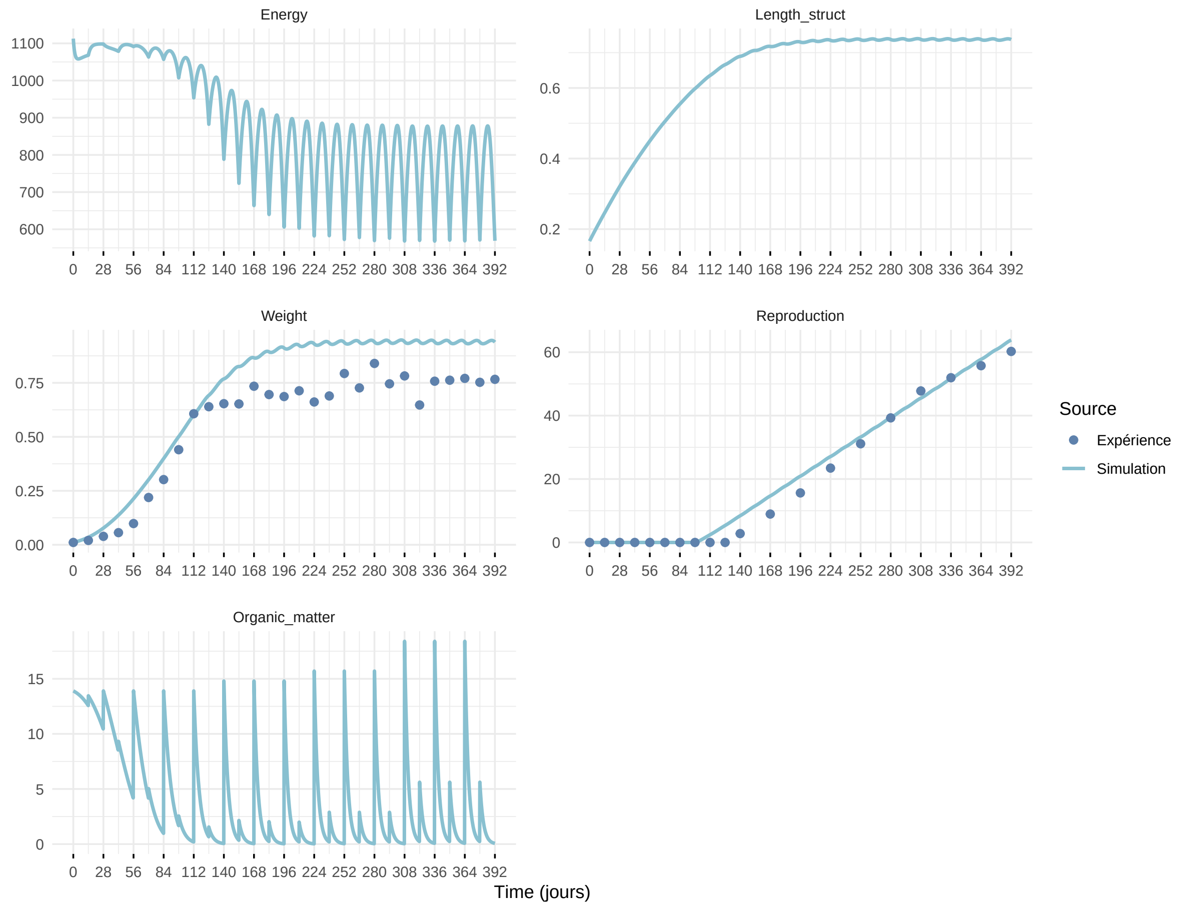



Expérience : Bart2019_XP3
Energy
Length_struct
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)

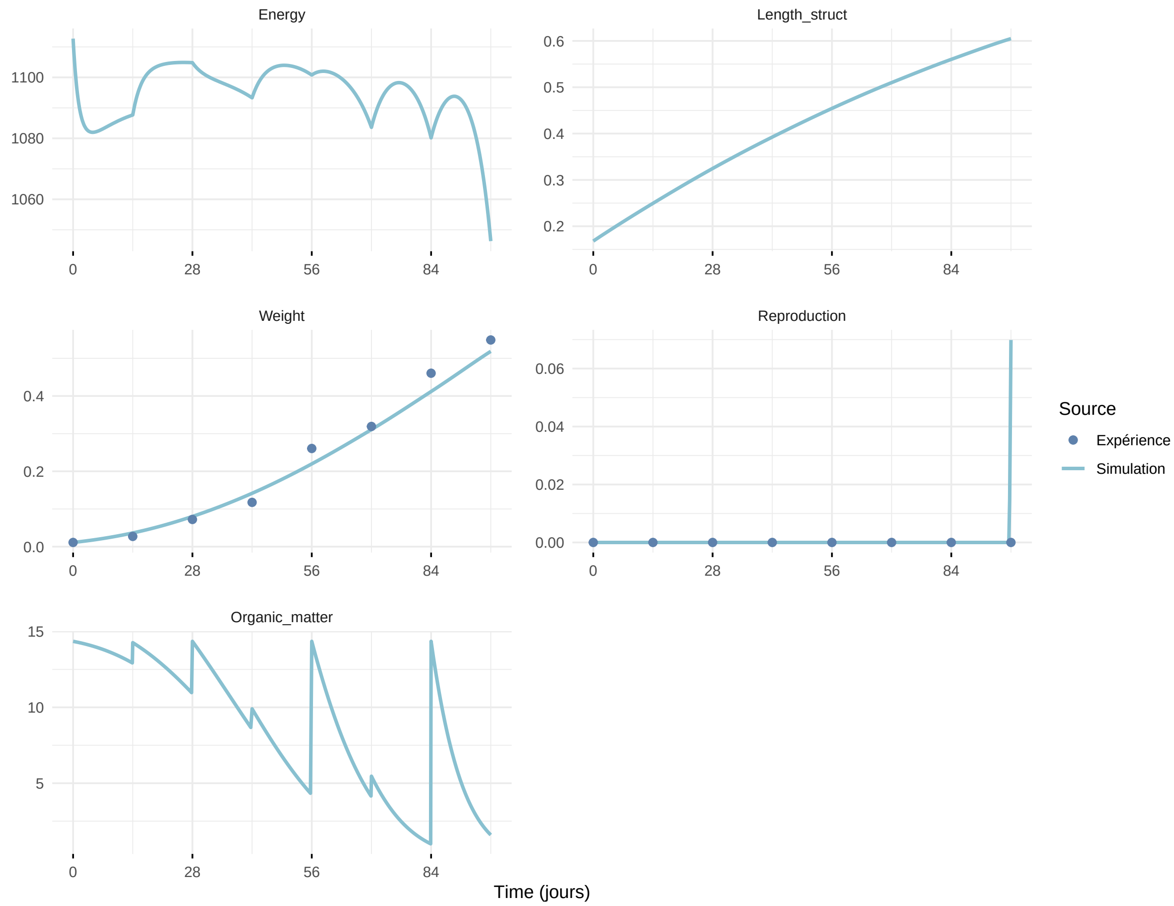
Expérience : Bart2019_XP4_1

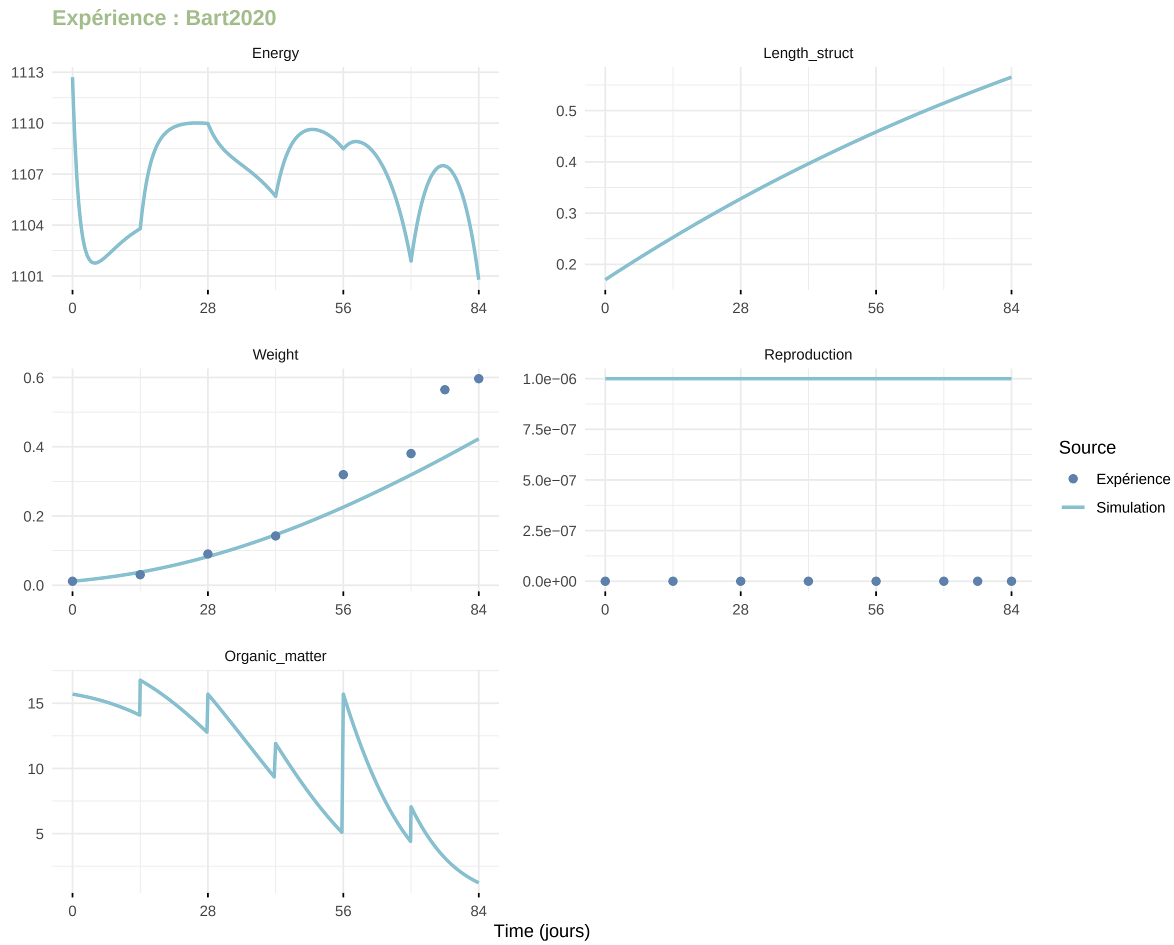


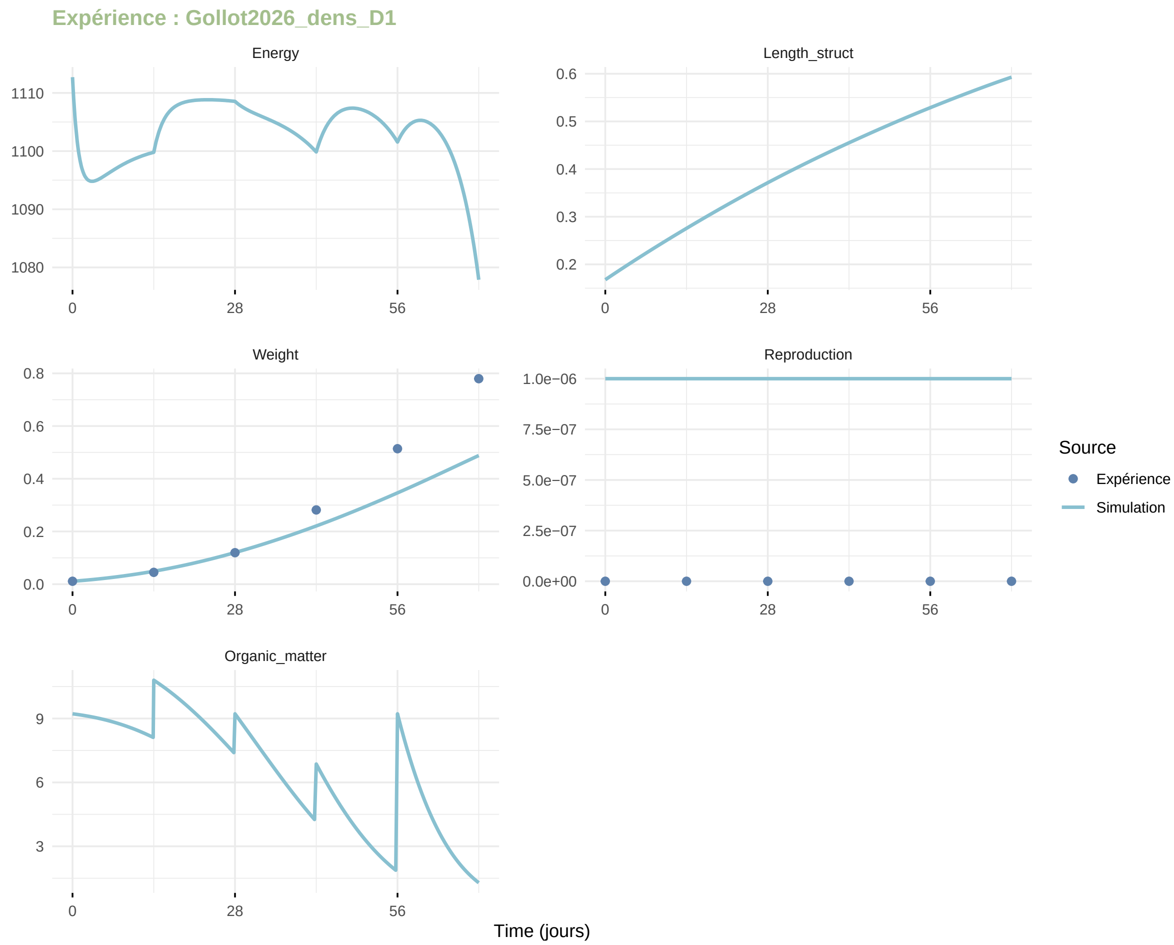
Expérience : Bart2019_XP4_2



Expérience : Bart2019_XP5

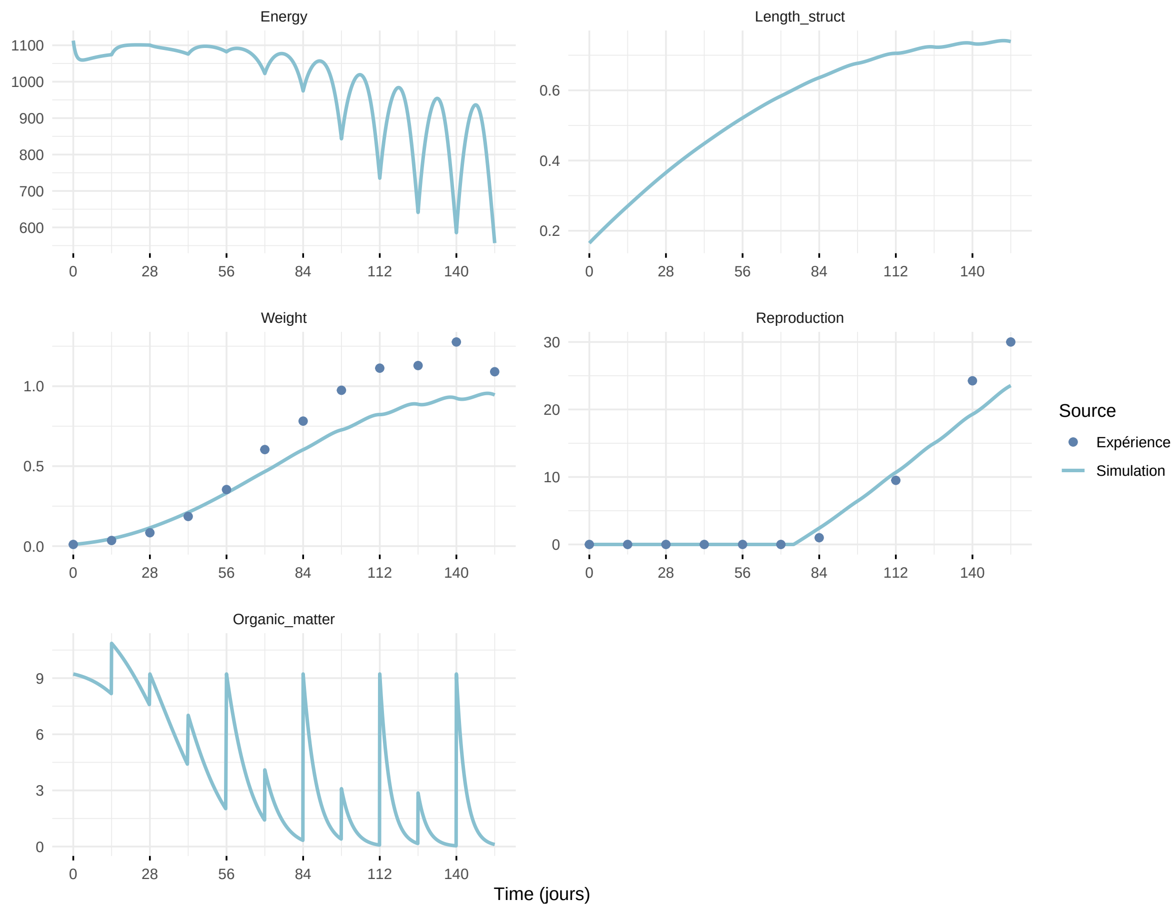



Expérience : Bart2020
Energy
1113
1110
1107
1104
1101
Length_struct
0.5
0.4
0.3
0.2
Weight
0.6
0.4
0.2
0.0
Reproduction
1.0e−06
7.5e−07
5.0e−07
2.5e−07
0.0e+00
Organic_matter
15
10
5
0
28
56
84
Source
Expérience
Simulation
Time (jours)

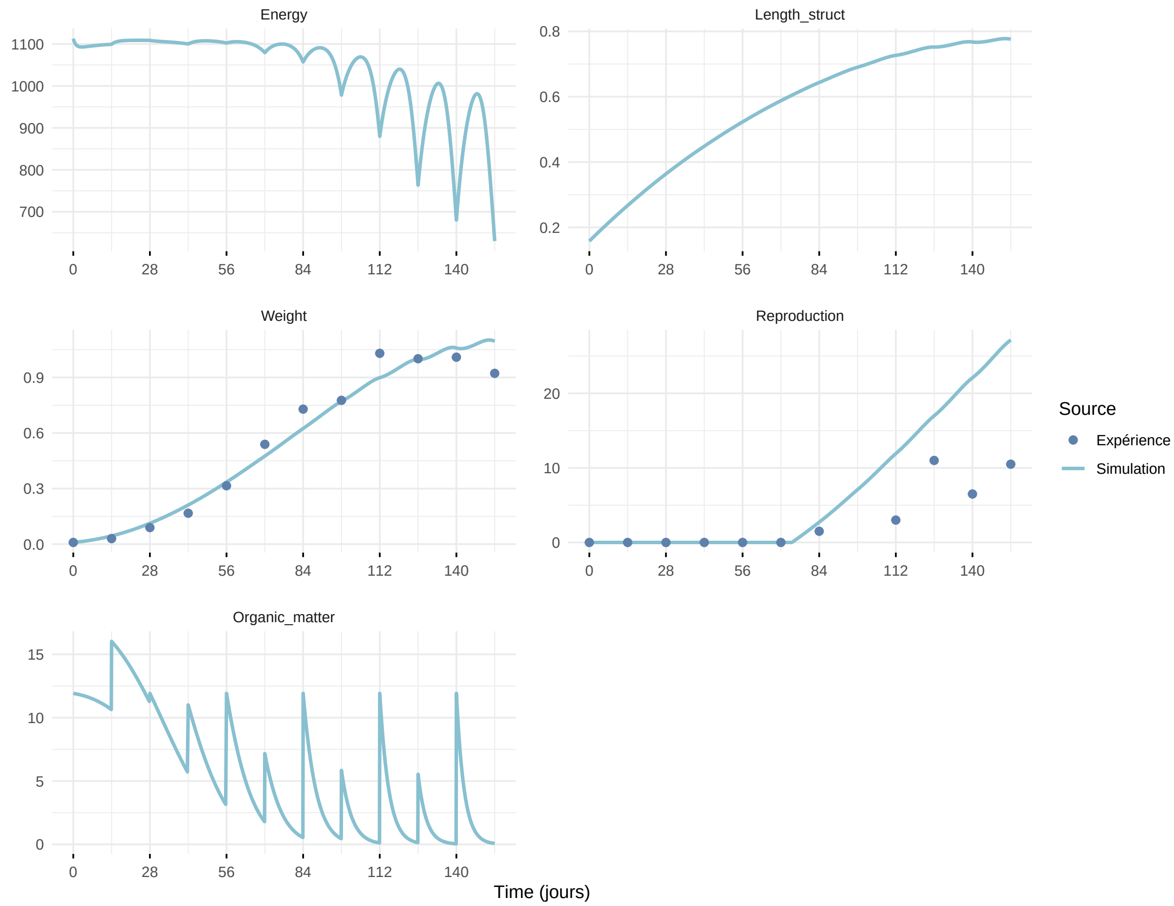


Expérience : Gollot2026_dens_D1
Energy
1110
1100
1090
1080
Length_struct
0.6
0.5
0.4
0.3
0.2
Weight
0.8
0.6
0.4
0.2
0.0
Reproduction
1.0e−06
7.5e−07
5.0e−07
2.5e−07
0.0e+00
Organic_matter
9
6
3
0
28
56
Time (jours)
Source
Expérience
Simulation

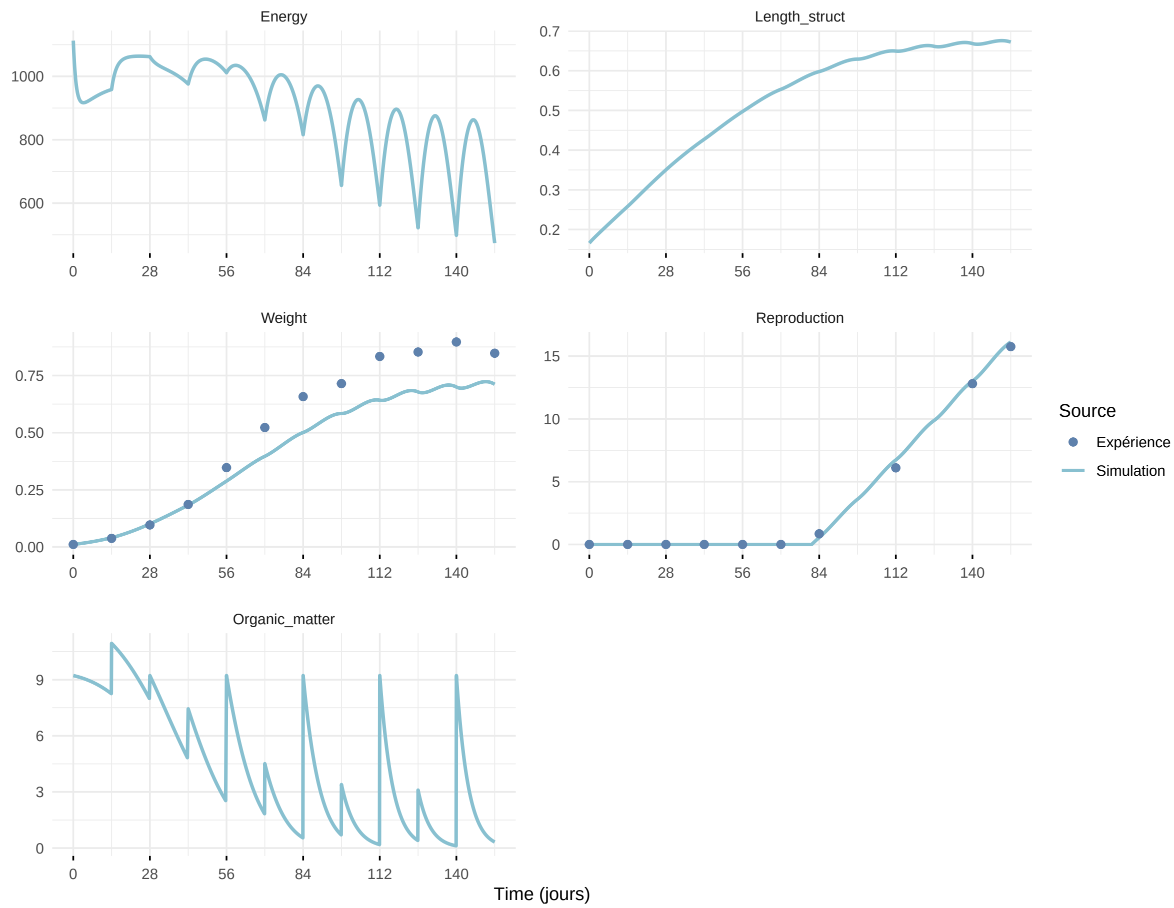
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7